

When Modal Contact Rows Fail in Fixed-Budget XPBD: Diagnosis and an Operating Guide

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Abstract

Rigid-body XPBD engines already resolve contacts and joints, and a small global modal state (16–24 amplitudes) is an attractive way to give a stiff prop low-cost vibration without a full nodal or tetrahedral deformable. We show that inserting this shared state directly into a two-way unilateral contact row need not keep modal storage within the measured rigid-side supply once the host runs only a few local iterations per substep. Across three scenes, two relaxations and four schedules, our tested XPBD implementation overdraws that supply by up to 4.4×10^7 J, augmented-Lagrangian and implicit implementations by at most 6.7 J in 3 of 24 cells and never, respectively. An iteration ladder, equal-row-count schedules and a within-host weight swap trace the amplification to the modal contact-row weight: the stiffness-aware implicit form removes the overdraw in every injecting cell while still retaining 0.70–0.89× the converged modal energy wherever the substep resolves the retained band, and where the weight is fixed the budget is better spent on iterations than on substeps or a block solve. For hosts that still cannot converge the row, a cumulative modal-storage projection enforces its accounting bound unconditionally (Prop. 4.1, roundoff floor in all 90 measured cells). Holding the contact-observed surface fixed, it cuts the penetration it opens from 2.4–47.6× the truncated solve’s own to 1.0–6.3× (both relaxations; Table 1 shows the 0.7 rows) without removing it: the truncated row opens the residue, which the projection still amplifies.

CCS Concepts

• Computing methodologies → Physical simulation.

Keywords

contact, modal reduction, position-based dynamics, energy safety, fixed-budget solvers

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1 Introduction

A developer building an interactive scene on a rigid-body XPBD engine [14] already has a solver resolving contacts and joints under a small, fixed local-iteration budget. To make a stiff prop such as a shelf, a table, or a ledge ring when struck, the heavyweight option is a full nodal or tetrahedral deformable with thousands of online coordinates; the lighter one is a precomputed modal basis of 16–24 global amplitudes $\mathbf{q}$ capturing the low-frequency bending of a stiff, small-displacement body [8, 9, 15]: a handful of extra runtime numbers, offline precomputation, one host stepping everything.

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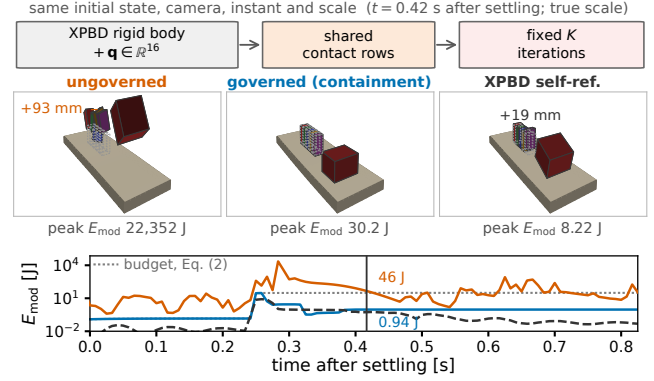


Figure 1: One shelf impact at a production-like interactive budget (1×8), true scale, one deterministic drop under three treatments. Ungoverned, truncation energy throws the resting books up to 93 mm, more than their own height; under the projection (5) the bound (2) holds, peak modal energy stays at 30.2 J and no book leaves rest. The bound also removes legitimate motion: the high-iteration reference (500×1) lifts one book 19 mm, the governed run none.

This paper is written for that developer, and does not argue XPBD against another solver.

The difficulty lies in the contact. In the real-time precedent the modal prop is driven *one way*: an external filter reads detected contact events and rings the modes with no back-reaction [3], double-counting the interface energy and leaving the prop’s deformation unable to change how it is supported. The *two-way* alternative resolves rigid and modal unknowns through one shared contact multiplier [20], so a heavier load bends the support and the bent support redistributes the load. But every support contact touches the same global $\mathbf{q}$, making the contact-row graph dense on the modes, and a host running a few local iterations per substep need not converge it. We take the row as given: the velocity-level law of Zheng and James [20] ($C^{n+1} \approx C^n + h \mathbf{J} \mathbf{u}^{n+1}$, same Jacobian and multiplier), narrowed to normal rows with $e=0$ and transcribed into the host’s positional-impulse idiom. That transcription is not neutral: Zheng and James [20] resolve against the implicit step’s reduced effective mass, whereas a position-based host weights the row by an inverse mass, which for mass-normalized modes is $\mathbf{I}$: the natural thing to write on this host, no contribution of ours, and the lever of §3.2. What does it cost at a normal budget, and what should one do about it?

Finite-iteration coupling across a partition boundary can inject spurious energy even when every subsystem is individually passive [18], and velocity-level frictional contact spanning rigid and reduced deformable bodies, with the energy artifacts of iterative solvers named, goes back at least to Kaufman et al. [10]. Missing

is a quantitative account of the present case, unilateral contact-to-modal transfer in a fixed-budget host, together with an operating guide; this paper supplies both.

Contributions.

- (1) **A practitioner workflow and its failure.** Treated as the workflow it is, adding a compact modal state to a fixed-budget XPBD host overdraws the measured rigid-side supply by up to 4.4×10^7 J at interactive budgets across three scenes, two relaxations and four schedules (§3.1), with augmented-Lagrangian and implicit implementations as milder controls, a gravity/no-contact accounting audit, and a deterministic perturbation ensemble.
- (2) **Mechanism and an operating guide.** An iteration ladder, equal-row-count schedules, warm starting, mode truncation and an in-host block-condensation ablation localize the amplification to the stiffness-blind modal weight in the shared row, exposed rather than caused by a truncated solve (§3.2), giving an ordered rule: a stiffness-aware weight where the architecture allows, otherwise iterations and a verified basis rather than more substeps or a block solve.
- (3) **A last-resort guardrail and its price.** A cumulative modal-storage bound funded by the measured rigid-side loss and enforced by a scalar projection of the realized state (§4): a deliberately minimal fail-safe, guaranteed unconditionally (Prop. 4.1, roundoff floor in all 90 measured cells), at a measured cost to contact validity (§4.1).

Related work. Rigid XPBD [14], interactive modal deformation and velocity-level modal contact [8–10, 20], and reduced XPBD [15] set the stage; the last uses modes as a deformation surrogate, whereas here modal amplitudes enter as contact DOFs driven by the shared multiplier (cf. penalty contact without a shared multiplier [21]). The guardrail adapts energy-tank control [5, 7, 16] rather than inventing a mechanism: it keeps the tank-and-observer structure but funds the tank from the measured rigid-side loss rather than from the port it regulates, and scales the realized state rather than a force, since a fixed-budget solver commits its multipliers before the excess is observable. That separates it from total-energy projection [4], bidirectional total-energy tracking [19], and the structural finite-budget passivity of bilateral couplings [18]. Two schemes are closer still: passive midpoint integration [11] attains passivity by modifying the integrator, whereas the host here is assumed fixed and its integrator and contact solve left untouched; and frame-rate energy clamping [17] caps energy per frame against a prescribed budget, whereas ours is cumulative and measured from the rigid-side loss, which lets a substep deposit more than it earns while the running total stays admissible.

2 The row and the invariant it must respect

Let $\mathbf{q} \in \mathbb{R}^r$ be mass-normalized modal amplitudes of the supporting body, so its modal mechanical energy is $E_{\text{mod}} = \frac{1}{2} \dot{\mathbf{q}}^\top \dot{\mathbf{q}} + \frac{1}{2} \mathbf{q}^\top \mathbf{K} \mathbf{q}$. A support-contact row between a rigid corner at world height y_c and the deformed surface has gap

$$C = y_c - (y_{\text{rest}} + \mathbf{U}_y^\top \mathbf{q}), \quad C \geq 0, \lambda \geq 0, C\lambda = 0, \quad (1)$$

with $\partial C / \partial \mathbf{q} = -\mathbf{U}_y$, so one multiplier λ drives both the rigid body and the modal amplitudes. Equation (1) is the transcription discussed above rather than a new law: one row, instantiated at every support contact in every substep (48 on the shelf, 40 on the ledge, 200 on the table), all sharing the one modal state $\mathbf{q}$.

The invariant. The energy stored in the modal subsystem must never exceed the measured gross rigid-side loss of (3) below, cumulatively over the run:

$$E_{\text{mod}}^n - E_{\text{mod}}^0 \leq \eta \sum_{k \leq n} \max(\Delta E_{\text{rig}}^k, 0), \quad \eta \in (0, 1]. \quad (2)$$

The transfer efficiency η sets how much of the measured loss may be spent; every experiment here runs $\eta = 1$, the least restrictive admissible value, so no result depends on tuning it. Being cumulative rather than per-step, the bound lets a substep deposit more than it earns provided the running total stays within budget, so a legitimate ring persists.

Definitions. Modal storage is the mechanical energy of the reduced state above, with $E_{\text{mod}}^0 = 0$: every scene starts from an undeformed, resting support, so even the run's own settling transient must be funded by the reservoir. The supply is

$$\Delta E_{\text{rig}}^k = (E_{\text{rig}}^{k,-} - E_{\text{rig}}^{k,+}) + W_g^k, \quad W_g^k = \sum_b m_b \mathbf{g}(\mathbf{x}_b^{k,+} - \mathbf{x}_b^{k,-}), \quad (3)$$

where $E_{\text{rig}} = \sum_b [\frac{1}{2} m_b \|\mathbf{v}_b\|^2 + \frac{1}{2} \omega_b^\top \mathbf{I}_b \omega_b]$ is purely kinetic (over all dynamic bodies, not only those in contact) and $\pm$ are the ends of substep k . We call this the **measured gross rigid-side loss**: a loss, not a dissipation, since it also counts legitimate rigid-to-modal transfer and rigid-rigid losses elsewhere in the scene, an envelope on what contact dissipated rather than a measurement of it (§6). Gravity enters only through W_g , so free ballistic motion registers zero supply and only an inelastic velocity change funds the reservoir; substep endpoints tile the timeline exactly, so no rigid energy change escapes the accounting.

Two diagnostics. The right-hand side of (2) is the measured loss of (3), never a tuned constant or a fraction of incident energy, and it constrains a scalar reservoir rather than a contact port, so the bound is neither port passivity nor a signed per-interface statement (§6 measures what its gross sum recycles). Two diagnostics stay distinct: the incident-energy ratio $R = \text{peak modal energy} / \text{peak incident rigid KE}$, a severity measure read once per frame, and the signed margin of (2) in joules, the invariant proper, evaluated against the running supply at every substep commit. They need not flag the same cells, and because the ledger sees intra-frame excursions the frame sample misses, a cell's margin may exceed its reported peak modal energy.

3 The failure at a fixed budget

All solver-behaviour measurements come from one machine (Apple M4, CPU, CPython 3.12), since chaotic contact stacks diverge across architectures under floating-point reassociation. Every number is frozen with its command and commit in the supplemental archive, which ships the per-cell CSVs and configurations behind every figure and table here, a code snapshot, a pinned environment and a smoke test; the supplementary video shows the comparisons at true scale.

3.1 One row, three implementations, the same 24 cells

We drive an impactor into a modally reduced support in three scenes (a shelf board, a ledge slab, and a 2.2×1.1 m table, the last two following the layouts of Coevoet et al. [3]) and sweep modal relaxation and the local budget (iterations $\times$ substeps) over $\{4 \times 1, 8 \times 2, 16 \times 4, 32 \times 8\}$. Per scene, as shelf / ledge / table: Young's modulus $0.5 / 10 / 1.1$ GPa; density $600 / 500 / 770$ kg m $^{-3}$; impactor $6 / 50 / 5$ kg dropped $0.50 / 0.80 / 0.50$ m; realized modal rank $16 / 16 / 24$, of which $6 / 4 / 14$ form a stiff cluster; retained modes spanning 20 Hz–24.7 kHz, 118 Hz–191 kHz, and 4.7 Hz–5.2 kHz. At the deployed h these give $\omega_{\max} h \approx 1.3 \times 10^3$, 1.0×10^4 and 2.7×10^2 , so (4) turns on: the two scenes that inject on the position-based host carry modes three to four orders above the substep rate; the table does not, though the augmented-Lagrangian host overdraws there, so (4) is necessary and not sufficient. The hosts are XPBD [12], AVBD [6] (from vertex block descent [2]), and a sequential-impulse realization in the sense of Catto [1], made implicit in the modal block, all pinned to a frame step $\Delta t = 1/120$ s subdivided into S substeps of $h = \Delta t / S$, so every cell advances the same physical time per frame and all step-dependent quantities are formed at h . They share per-substep contact regeneration, Rayleigh damping, an implicit-midpoint modal restoring step and $\eta = 1$, and differ in the unknown solved for (position, position + multiplier, or velocity), the modal contact weight, warm start and relaxation (full table in the supplement). Relaxation is inert on the impulse host (its implicit weight needs none), so its two rows are bit-identical, and warm start is as-deployed ($\lambda \leftarrow 0$ every substep for the position-based host, carried duals for the others) and not the lever: warm-starting it leaves the footprint unchanged (the same $8/24$ cells, worst 1.20×10^5) and is worse, not better, at 4×1 (Fig. 3a). The decisive difference is the contact-row weight, isolated with a within-host control in §3.2.

Equal row evaluations, unequal outcome. A cell costs $K \cdot S$ row evaluations per frame, so the budget axis conflates two refinements that are neither interchangeable nor equally costly. At 32 row evaluations on the shelf drop, spent as iterations ($K=32, S=1$) the ratio is 0.300 and (2) holds; spent as substeps ($K=4, S=8$) it is 3.13, over-drawing by 481 J. Substeps do reduce the ratio (Fig. 3a), roughly 2200 $\times$ over the same range, but in this equal-row shelf comparison they are the less cost-effective refinement; we claim the inversion for this row, not for constraints generally.

Figure 2 shows the outcome. The position-based host exceeds $R = 1$ in 8 of 24 cells, worst 1.2×10^5 (ledge, relaxation 1.0, 4×1 : peak modal energy 4.44×10^7 J against an incident 371 J), the augmented-Lagrangian host in 2 of 24 (worst 1.70), so neither is unconditionally benign; the ordering also inverts with the scene, the augmented-Lagrangian host being the worse on the table (1.70 against 0.37).

The ratio is not the invariant. We also measure (2) directly on all three hosts, ungoverned (the accounting is forced inert; trajectories are bit-identical). The diagnostics disagree: the position-based host violates it in 9 of 24 against $R > 1$ in 8, the augmented-Lagrangian host in 3 against 2, the implicit host in neither (0/24, at the roundoff floor). Magnitude orders the failures: the position-based overdraw

reaches 4.4×10^7 J, seven orders beyond the largest augmented-Lagrangian one (6.7 J, at the starved 4×1 table corner) and the implicit host's exact zero. A resting-contact, no-impact audit puts the augmented-Lagrangian and implicit control floor below 10^{-3} J, so that overdraw is real impact-driven injection rather than accounting noise. The position-based host has no small floor: at a starved budget it already injects 10^3 J from resting settling alone, the same failure without an impact event rather than a separate one.

3.2 The position-based amplification is a truncation artifact

Figure 3 isolates the local iteration budget. Two references recur below: the *implicit reference*, the implicit realization run to $K=500$ (the same code path, only more iterations), and the *XPBD self-reference*, the position-based host run to $K=500$ against itself. The position-based host falls monotonically from 2.96×10^4 at $K=1$ to 0.300 at $K=32$, approaching the implicit reference 0.2735, the gap $|\text{XPBD} - \text{reference}|$ shrinking from 2.96×10^4 to 2.7×10^{-2} , while the implicit realization moves only in the fourth decimal across $K = 2 \dots 500$ (max–min 6.8×10^{-4}).

A falling energy ratio could shrink coincidentally; measuring the constraint directly resolves that, with the two complementarity conditions read in their own units. End-of-substep penetration falls from 27.9 mm at $K=1$ to $3.6 \mu\text{m}$ by $K=64$, where it saturates, while the multiplier surviving on separated rows (force where the gap is open, in units of the steady-state median) rises from $84\times$ to $139\times$ through $K=24$ and collapses to zero only at $K=64$. The energy crossing at $K \approx 24$ therefore coincides with the gap converging, not with complementarity holding: the multiplier side clears a rung later. These measurements concern the position-based host only, the implicit realization being converged at $K=2$.

Self-convergence stops short of the reference. Running the position-based host to $K = 500$ against itself does not close the gap: its ratio plateaus by $K \approx 64$ at 0.2996, a fixed point 2.6×10^{-2} above the reference's 0.2735, not a tail. The state agrees no better: the deflection peak matches to 2.4%, the trajectories differing by 6.8 mm (34% of the reference's peak). The two hosts converge to different discrete solutions, as expected of two discretizations of one law, and the sweep separates a truncation pathology that convergence removes from a discretization difference it does not.

The contact-row weight is the lever. The three-host comparison of §3.1 confounds solver family with the modal contact-row weight; a within-host control separates them. Three quantities must be kept apart: the *compliance* $\tilde{\alpha} = \alpha / h^2$, XPBD's regularizer added to the row denominator (the support row is near-rigid, $\alpha_{\text{sup}} = 10^{-8}$, and the symplectic step's separate modal-elastic constraint carries its own $\tilde{\alpha}_e$); the *modal inverse mass* $\mathbf{W}_q$ [kg $^{-1}$] multiplying the Jacobian; and the *modal mobility* $\mathbf{U}_y^T \mathbf{W}_q \mathbf{U}_y$, the modal share of that denominator. The lever is $\mathbf{W}_q$, not either compliance.

With mass-normalized modes the deployed row uses the *true* $\mathbf{W}_q = \mathbf{M}_q^{-1} = \mathbf{I}$, so its mobility is independent of ω_i : the row is stiffness-blind. The stiffness-aware alternative uses the effective inverse inertia $\mathbf{W}_q = (\mathbf{M}_q + h \mathbf{D}_q + h^2 \mathbf{K}_q)^{-1}$, per mode $1 / (1 + h d_i + (\omega_i h)^2)$. Holding the position-based host fixed in every other respect (position-level unknown, warm start, relaxation, per-substep contact regeneration,

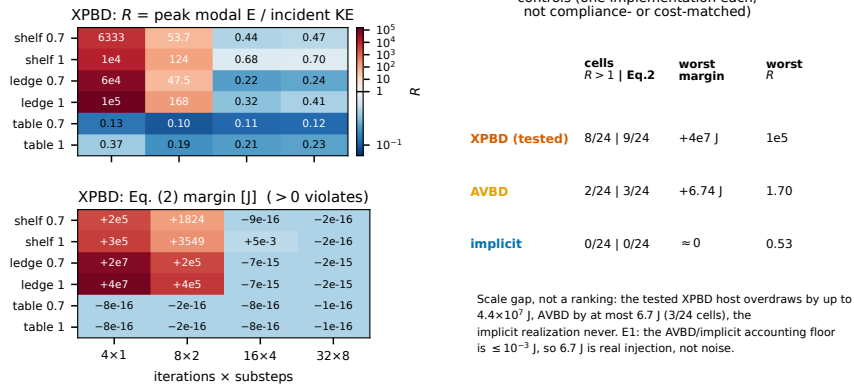


Figure 2: Our tested XPBD implementation over the 24-cell sweep (3 scenes × relaxation {0.7, 1.0} × budget), governor OFF. Left: $R = \text{peak modal energy} / \text{peak incident rigid KE}$ (top) and the signed margin of (2) in joules (bottom; > 0 violates); the two need not flag the same cells (8/24 vs 9/24), itself a result. Right: a control strip for the three implementations (one of each, neither compliance- nor cost-matched): a scale gap rather than a ranking, 4.4×10^7 J vs. ≤ 6.7 J vs. 0, with an accounting audit putting the control floor below 10^{-3} J, so 6.7 J is real injection (full 72-cell heatmap in the supplement).

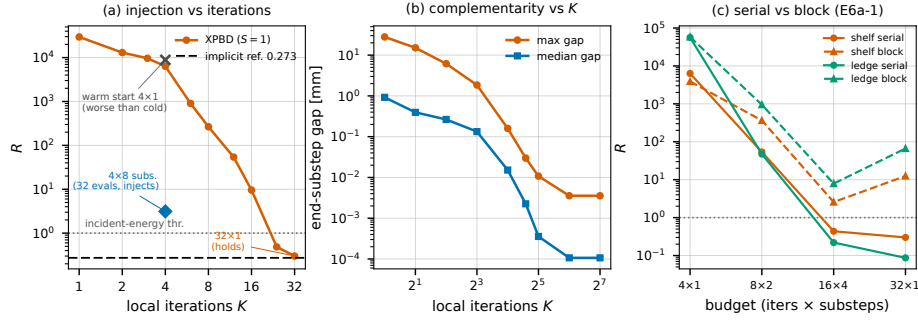


Figure 3: The XPBD operating envelope (our tested position-based host, governor OFF). (a) Injection R vs. local iterations K on a deterministic shelf drop ($S=1$): R decays monotonically toward the implicit reference (0.273), crossing the incident-energy threshold between $K=16$ and 24; annotations mark the 32×1 -vs- 4×8 equal-row pair and warm starting. (b) The end-of-substep gap converges with K , so the falling ratio tracks the row converging rather than one scalar shrinking. (c) Serial per-row support projection (the path used here) vs. joint block condensation (supplement E6a-1), both scenes: the serial path holds by 16×4 – 32×1 , condensation injects where it holds.

symplectic modal step) and replacing only $\mathbf{W}_q$ removes the overdraw in every injecting cell: $8/24 \rightarrow 0/24$, the deployed ledge 4×1 falling from $R = 1.2 \times 10^5$ to 8×10^{-5} and never worse at any budget (the baseline arm reproduces Fig. 2 exactly).

The swap contains the injection without simply deleting the response, provided the substep resolves the retained band. At the injecting shelf 8×2 cell it stores 5.57 J against the implicit reference's 7.92 J (0.70×, where the deployed row stores 1555.6 J), and across the shelf and table at 16×4 – 32×8 it holds 0.70–0.89× the deployed row's own converged energy. The ledge is the exception that proves the rule: at $\omega_{\max} h \approx 10^4$ the same weight damps even its softest retained mode to 2.5% mobility, so there a stiffness-aware weight and a step-aware basis are not separable choices. The mechanism is their ratio: the deployed row over-estimates how freely mode i

moves within a substep by

$$[\mathbf{W}_q^{\text{depl}}]_{ii} / [\mathbf{W}_q^{\text{impl}}]_{ii} = 1 + h d_i + (\omega_i h)^2 \approx (\omega_i h)^2 \quad (\omega_i h \gg 1), \quad (4)$$

so a stiff mode is displaced as if it had no restoring stiffness, and the restoring step values that over-displacement at $\frac{1}{2} \omega_i^2 \Delta q_i^2$. On the ledge ($\omega_{\max} h \approx 10^4$, §3.1) (4) reaches $\sim 10^8$ on the stiffest retained mode. The failure is thus a property of a stiffness-blind row weight at $\omega_i h \gg 1$ rather than of the fixed budget as such, consistent with the iteration ladder, the equal-row comparison, and the implicit host's convergence at $K=2$.

Convergence therefore largely cures the amplification (more iterations, the implicit realization, or merely its contact-row weight), so the bound of §4 is no substitute for it and targets only the regime where neither escape is available. That regime is common: interactive position-based solvers run near 1×8 – 2×4 [13], well below the $K \geq 24$ this scene needs, and there the position-based host violates

(2) in all six deployed cells (worst $R = 2282$, on 8 row evaluations) while the others match the sweep (1/6, 0/6).

Condensing the shared rows is not the fix. What converges the row is the iteration budget, not how the shared rows are scheduled: replacing the serial per-row support projection with a joint block condensation (modal weight, restoring step, budget and warm start fixed; Fig. 3(c)) does not remove the injection. It is comparable at the starved 4×1 corner and markedly worse wherever the serial path holds ($R = 12.5$ and 66.5 at 32×1 on shelf and ledge, against 0.30 and 0.09 ; the budget axis is ordered by schedule, not by the $K \cdot S$ cost, so 32×1 is half the cost of 16×4). In the two scenes and four budgets tested, condensation exposes a worse convergence path, not a cure; we do not claim this of block solves in general.

Robustness of the failure. The catastrophic cells are not a knife-edge. For each worst cell, the equal-row pair and the guardrail cell we run 16 paired deterministic perturbations (12 physical, 4 contact-row-order permutations; supplement). Every injecting cell overdraws in all 16, the sign never flips, and the 4.4×10^7 J anchor sits near the top of a $2.2\text{--}4.5 \times 10^7$ J band; this establishes local insensitivity, not generalization. Mode truncation localizes the amplification without removing it: excluding the stiff cluster cuts the shelf 4×1 ratio $6333 \rightarrow 22.1$, yet 6 of 8 injecting cells still overdraw (worst ledge $+1.24 \times 10^6$ J), so band-limiting is a first-line mitigation rather than a fix.

4 A last-resort guardrail

When a fixed host cannot converge the row and neither the implicit realization nor more iterations are available, a deliberately minimal fail-safe holds (2) regardless of the contact solve.

Enforcement. A reservoir B accumulates the supply and is debited by realized storage, both after the contact solve within the same substep, since ΔE_{rig} is not observable until the velocity solve finalizes the rigid velocities. The substep credits $B \leftarrow B + \eta \max(\Delta E_{\text{rig}}, 0)$, tests realized modal energy against $\bar{E} = E_{\text{mod}}^- + B$, projects into $\{E_{\text{mod}} \leq \bar{E}\}$ if it would overdraw, then debits $B \leftarrow \max(B - \max(\Delta E_{\text{mod}}, 0), 0)$ ($E_{\text{mod}}^\pm$ are the modal energies before and after the solve).

Which admissible point is free, and the choice matters: scaling the whole state shrinks the load-bearing sag $\mathbf{U}_y^\top \mathbf{q}$ that resting bodies stand on, lifting the support into them (§4.1), so we project along directions the active rows cannot see. Let $\mathbf{U}_c$ collect the support rows whose gap has closed, $\mathbf{d} = \mathbf{U}_c \mathbf{q}$, and $\mathbf{q}_{\text{qs}} = \mathbf{K}^{-1} \mathbf{U}_c^\top \mathbf{y}$ the least-elastic-energy state with the same $\mathbf{d}$, where $(\mathbf{U}_c \mathbf{K}^{-1} \mathbf{U}_c^\top) \mathbf{y} = \mathbf{d}$ (pseudo-inverted where the engaged rows outnumber r) and $E_{\text{qs}} = \frac{1}{2} \mathbf{d}^\top \mathbf{y}$. The remainder $\mathbf{q}_\perp = \mathbf{q} - \mathbf{q}_{\text{qs}}$ satisfies $\mathbf{U}_c \mathbf{q}_\perp = \mathbf{0}$ and $\mathbf{q}_{\text{qs}}^\top \mathbf{K} \mathbf{q}_\perp = 0$, so

$$(\mathbf{q}, \dot{\mathbf{q}}) \leftarrow (\mathbf{q}_{\text{qs}} + s \mathbf{q}_\perp, s \dot{\mathbf{q}}), \quad s = \min\left(1, \sqrt{\frac{\bar{E} - E_{\text{qs}}}{E_{\text{mod}}^+ - E_{\text{qs}}}}\right) \quad (5)$$

leaves $\mathbf{U}_c \mathbf{q}$ exactly unchanged while landing on $\bar{E}$. If $E_{\text{qs}} > \bar{E}$ the observed surface is itself unaffordable: the deepest-engaged prefix of rows that does fit is preserved instead (a bisection, E_{qs} being monotone in the row set), failing which the state falls back to the

Table 1: Where the penetration comes from: worst end-of-substep gap violation, position-based host, relaxation 0.7, measured identically in three arms. *unclamped* = governor off (the truncated solve’s own penetration), *scaled* = whole-state projection, *preserved* = (5). Clamp count, impulse (next substep’s normal impulse over the unclamped steady-state median) and λ variance (over the engaged support rows, relative to the unclamped arm) are the preserving arm’s; 4×1 clamps 83–96% of substeps, leaving no steady state (n/a).

scene	budget	clamped	worst penetration [mm]			impulse	λ var.
			unclamped	scaled	preserved		
shelf	4×1	104/108	6.16	21.59	8.31	n/a	n/a
shelf	8×2	201/216	0.50	6.43	0.50	$6.9 \times$	$50 \times$
ledge	4×1	90/108	8.80	21.19	17.79	n/a	n/a
ledge	8×2	166/216	0.91	4.08	0.93	$8.8 \times$	$61 \times$

always-feasible ($\beta \mathbf{q}_{\text{qs}}, \mathbf{0}$), $\beta = \sqrt{\bar{E}/E_{\text{qs}}}$. With no active row, $\mathbf{q}_{\text{qs}} = \mathbf{0}$ and (5) is the whole-state scale $\gamma = \sqrt{\bar{E}/E_{\text{mod}}^+}$.

All three hosts run this loop identically once ΔE_{rig} is formed from (3). Contact is not re-solved and the active set is read from geometry alone, so no host-specific multiplier is needed, and the projection is inert within budget ($s=1$, bit-identical trajectories).

PROPOSITION 4.1. *For any $\eta \in (0, 1]$ and any host, the loop maintains $B \geq 0$ and guarantees (2) unconditionally, assuming nothing about the contact solve beyond the projection being the last write to $(\mathbf{q}, \dot{\mathbf{q}})$ in the substep, up to an accounting tolerance ϵ over n substeps (ϵ the per-test tolerance, 10^{-12} J).*

PROOF SKETCH. Carry two invariants over substeps, with $c_k = \eta \max(\Delta E_{\text{rig}}^k, 0)$: $B_k \geq 0$ and $(E_{\text{mod}}^k - E_{\text{mod}}^0) + B_k \leq \sum_{j \leq k} c_j$ (both hold at $k=0$). Crediting c_k before the test caps realized storage at $E_{\text{mod}}^- + B_{k-1} + c_k$, so $\Delta E_{\text{mod}} \leq B_{k-1} + c_k$; the debit keeps $B_k \geq 0$, and both invariants follow by induction. Since $B_n \geq 0$, this is (2) itself rather than a relaxation, the same-substep credit absorbing the granularity term a naive telescoping would leave. The argument uses only that the projected state lands in $\{E_{\text{mod}} \leq \bar{E}\}$, never the direction it moves, leaving (5) free to preserve the surface. $\square$

The governor holds. With the governor on, all 72 sweep executions and the 18 deployed cells of §3.2 satisfy (2): 90 governed cells (78 distinct, the implicit relaxation pairs being bit-identical) at the roundoff floor (worst margin 3.6×10^{-15} J). Preserving the surface stores what the bound permits rather than the least it can, so the worst realized ratio is 1.15 (position-based) and 1.70 (augmented-Lagrangian) against 1.21 and 1.31 scaled, the invariant holding in both.

4.1 The cost of enforcement

Accuracy. Bounding the energy helps only if the bounded solution improves on the unbounded one. At a moderate injecting cell (shelf, 8×2 , relaxation 0.7) peak modal energy is 1555.6 J un-governed and 28.7 J governed against the implicit reference’s 7.92 J, an energy error falling $196 \times \rightarrow 3.6 \times$. The trajectory still degrades, though far less than under a whole-state scale: $\|d - d^{\text{ref}}\|_\infty$ rises $6.5 \rightarrow 9.2$ mm ($33 \rightarrow 46\%$ of the reference peak) against 14.2 mm

(71%) scaled. The governor is a safety envelope, not an accuracy device.

This is also why a 196× energy error coexists with a peak deflection only 1.2× too large: 99.6% of the excess sits in the stiff mode cluster above 20 kHz (83× this cell's 240 Hz substep rate). A whole-state scale cannot separate that from the legitimate sag (99.6 → 97.7% above the cut), whereas projecting along the contact-invisible directions removes proportionally more (99.6 → 89.8%).

Contact validity. Any projection that rewrites $\mathbf{q}$ after the solve moves the support surface without re-solving contact, so it can open a gap violation. Table 1 separates the contributions, instrumented by wrappers that leave the solve unperturbed (clamp counts reproduce the independent sweep exactly).

The truncated row opens the penetration; the guardrail amplifies it. At converged budgets no substep clamps and the worst violation is 0.011–0.124 mm (16×4, 32×1, both scenes), so the artifact exists only where the row is truncated; at starved budgets the unclamped solve already opens 0.17–8.8 mm with no governor present. The projection then *amplifies* that residue: a whole-state scale by 2.4–47.6× (21.6 mm, 72% of the 30 mm board), preserving the observed surface only 1.0–6.3×. The exception is ledge 4×1 (17.8 mm), where the observed surface is itself the injection artifact that no bound-respecting projection can keep. Nor does the corrective transient soften: the next substep's normal impulse is 6.9–8.8× steady state, so preservation buys fewer millimetres, not a gentler recovery (supplement).

5 Fidelity and runtime

The reduced response against a full-FEM reference. Modes are worth using only if the reduced response is faithful. Against an unreduced FEM solve on the same discrete operator (so discretization bias cancels), the reduced ledge response tracks the reference as the step refines, but not at the step we deploy: the peak-deflection ratio rises 0.38 → 0.54 → 0.79 → 0.88 as Δt refines 1/120 → 1/960 at fixed $S=4$, and ring frequency agrees to 0.4% (78.0 vs 78.3 Hz, the loaded ring rather than a basis frequency) only at 1/960, the deployed 1/120 shifting the recovered ring to 47.25 Hz. At the operating point of every other experiment here the reduced model therefore captures 38% of the reference peak, the residual at 1/960 being ~10% $k=24$ truncation rather than coupling error, and the far-field falloff tracks at Spearman $\rho = 0.89$ with no fitted attenuation. This ladder uses a richer $k=24$ ledge basis than the rank-16 ledge of the sweep (§3.1), so it bounds fidelity from above. The limit is a property of the reduction at interactive steps, not of the injection diagnosed here.

Runtime cost. On the CPU host at 16×4 the ledger adds 0.23–0.41 ms on the shelf and ledge, 0.9–3.4% of those scenes' 11–31 ms steps; on the table it does not resolve above run-to-run noise (+1.35 ± 2.08 ms over 10 repetitions), so we bound it by ~1.4 ms instead, plus one $m \times m$ solve per clamped substep ($m \leq 8$ here). At the tight deployed 1×8/2×4 schedules it costs 0.3–2.2 ms, or 6.6–34.3%, so the guardrail is proportionally most expensive in the regime that needs it. These are Python/CPU float64 steps and the percentages are ratios within this prototype rather than a frame budget: we make no unqualified real-time claim (§6).

Adding modal DOFs to a fixed-budget XPBD host?

a diagnostic + operating guide — not a solver ranking, not a governor method

1 Can you change the contact row?

Make its modal weight stiffness-aware:
implicit $(M+hD+h^2K)^{-1}$, not $1/M$.
within-host control: 8/24→0/24 overdraws, no host swap

2 Stuck with the XPBD shared row?

Audit it: prefer iterations over substeps;
band-limit the basis.
32×1 holds ($R=0.30$); 4×8 overdraws (+481 J);
band-limiting still leaves 6/8 cells over

3 Need a hard energy guarantee?

The cumulative storage bound contains
modal energy — last resort.
holds in 90/90 cells, but still leaves up to 17.8 mm
penetration + a worse trajectory. Containment, not a fix.

The governor is a safety net, not the product.

Figure 4: The operating guide as an ordered decision (§3.1, §3.2, §4); the guardrail is the last resort, not the product.

6 Limitations

Stability is not accuracy. Where the projection is load-bearing it still opens contact violations (up to 17.8 mm) and moves the trajectory away from the implicit reference even as it moves the energy toward it (§4.1): bounded, not faithful. Preserving the observed surface narrows that gap without closing it and leaves the corrective transient unreduced; a load-weighted subspace and one-sided constraints remain unbuilt.

Scope of the row. The coupling is normal-only (friction stays body-local, so scraping and rolling lie outside it) and fixes $e=0$. The bound governs modal storage, not spurious energy in rigid DOFs: in the impulse realization box–box rows rectify a support ring into net rigid motion, to which the ledger is blind by design.

Supply and coverage. The supply is a gross sum, so energy returned to the rigid bodies and re-dissipated is credited twice; the opposite-channel exposure $\sum_k \max(-\Delta E_{\text{rig}}^k, 0)$ reaches 3–32% of supply (position-based), 0.4–27% (implicit) and 102–118% on the augmented-Lagrangian host, whose reservoir is therefore funded largely by oscillation rather than dissipation. It is also uncapped, so (2) bounds a run, not a limit. Enforcement is host-side and single-machine (CPU float64), and with one implementation of each formulation in one contact regime, any claim that descent-class solvers are passive here is empirical about this sweep rather than a theorem.

7 Conclusion

For a developer adding low-cost vibration to a stiff prop on a fixed-budget XPBD host, the decisions are ordered (Fig. 4); the real question is not XPBD versus another solver but the contact-row weight. Where the two-way row overdraws, make that weight stiffness-aware (4) or spend the budget on iterations; only a truly fixed host needs the cumulative bound, which contains rather than corrects. Modal state is cheap; a truncated solve is not.

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